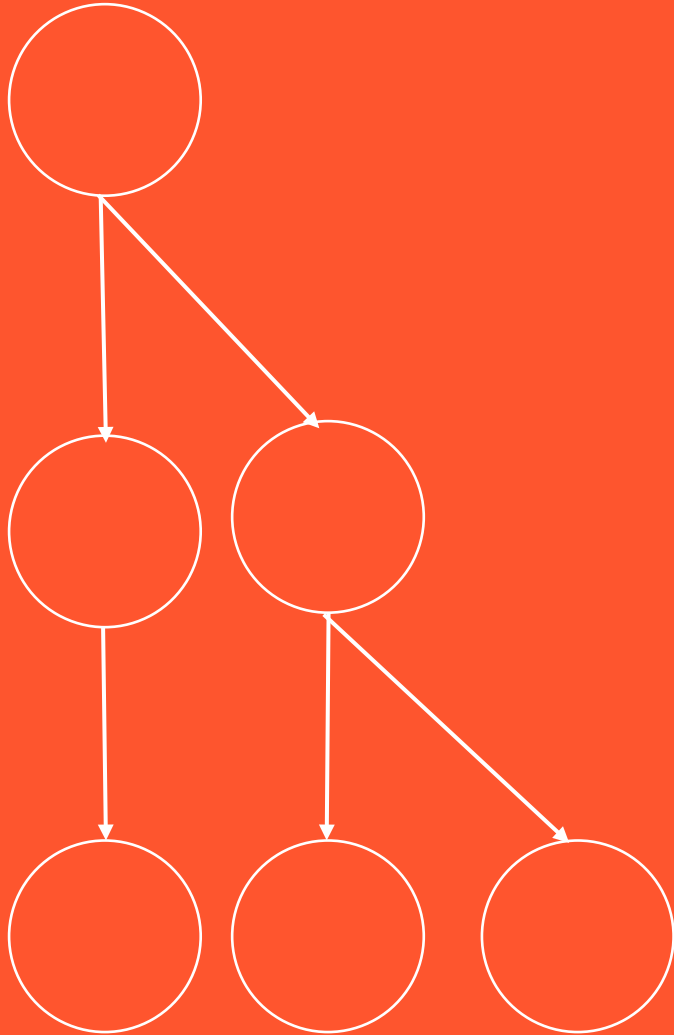


BCOG 100

Social Cognition

Lecture 3: Primate Brains



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 10-3

Password: pfc

Question: In the context of this week's topic, what is a false belief task, and how do scientists use it?

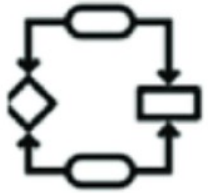


Primate Brains



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

Last Time:

Last time we identified three capacities underlying new primate behaviors: 1) modeling other minds, 2) modeling their own minds, 3) observational and learning

Today:

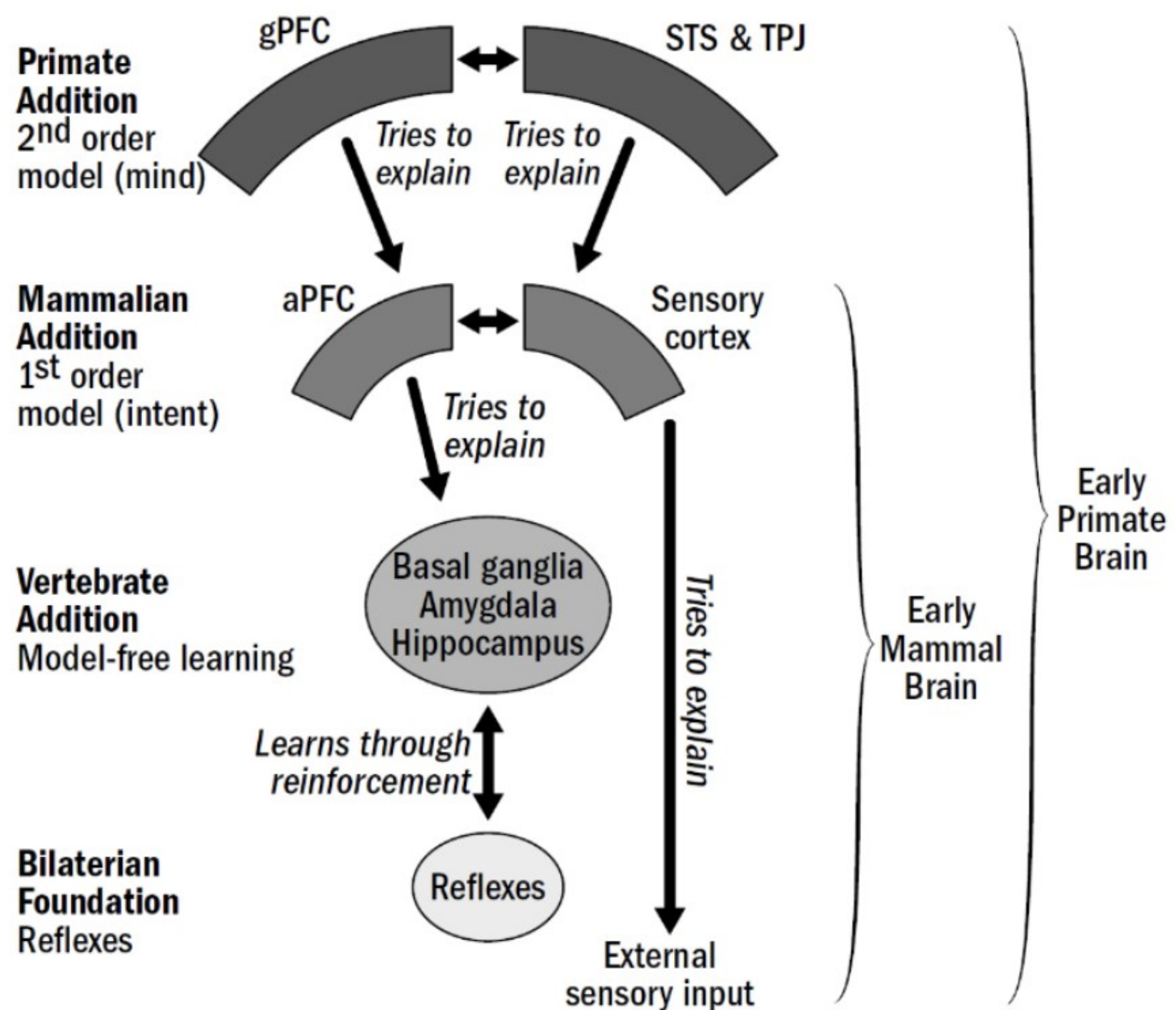
Today we ask what internal capacities made them possible

Source: Tartarelli, G., Bisconti, M. Trajectories and Constraints in Brain Evolution in Primates and Cetacean
Human Evolution 21, 275–287 (2006). <https://doi.org/10.1007/s11598-006-9027-4>

Primate Brains

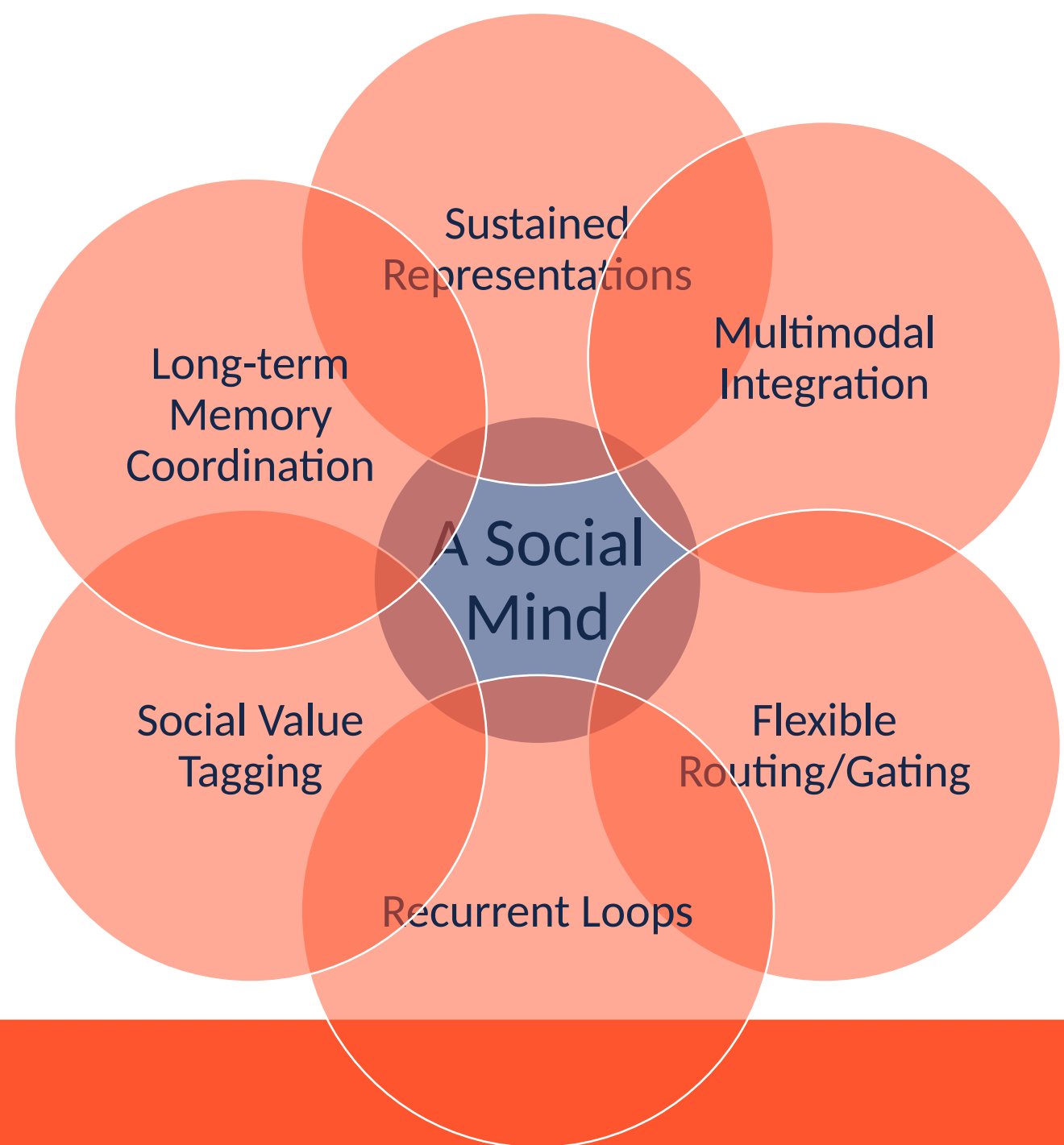
Primate Brains Were Not *Just* Bigger

- Not only size
- New cortex, new wiring
- New principles of control



Primate Brains

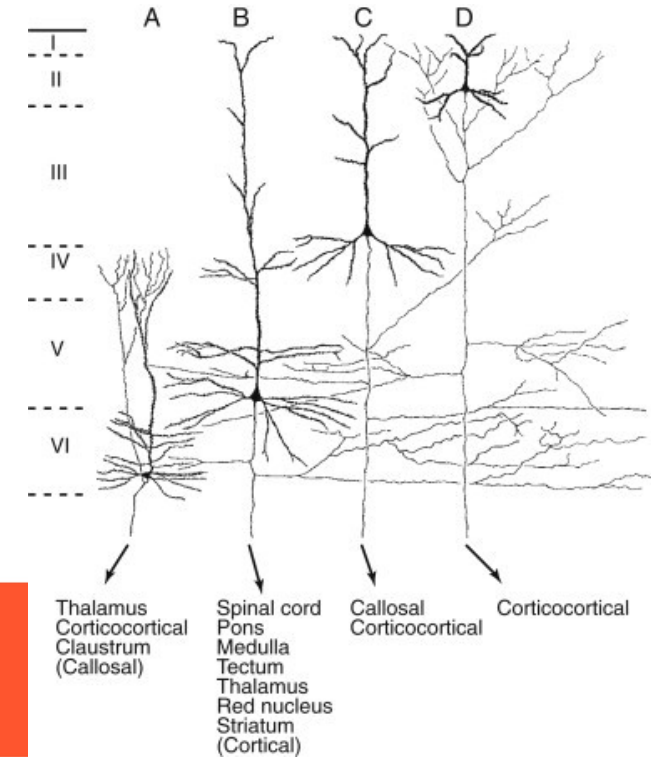
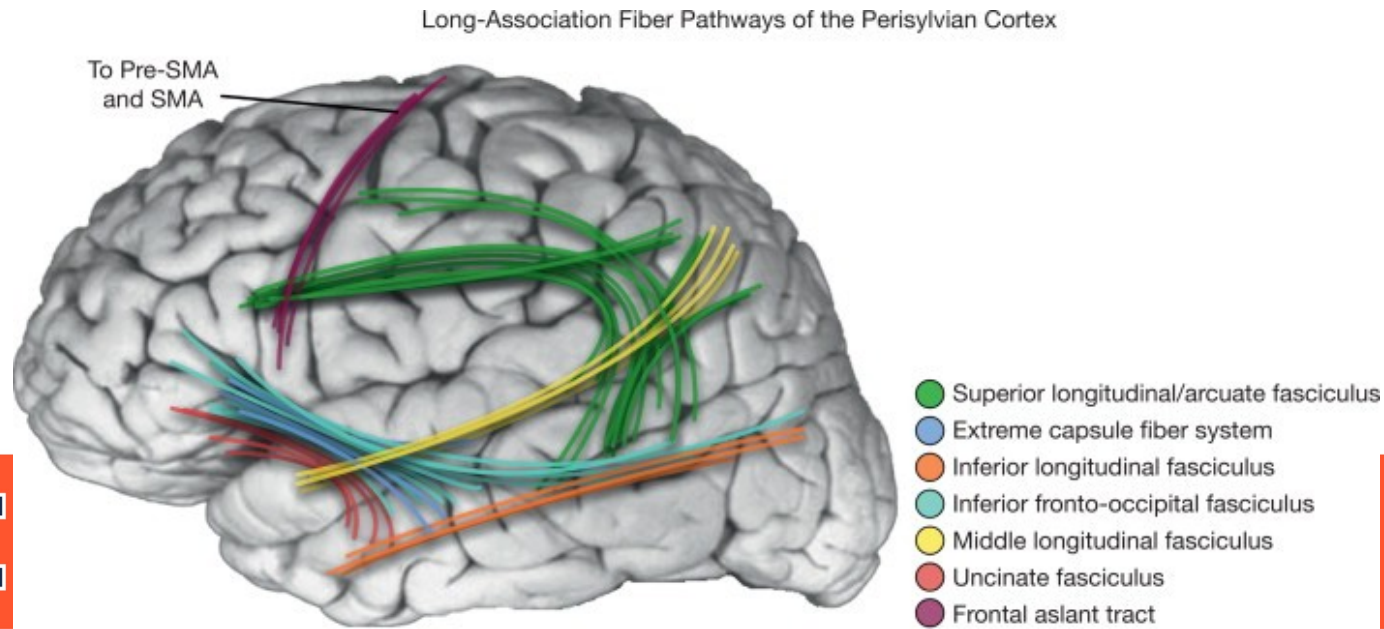
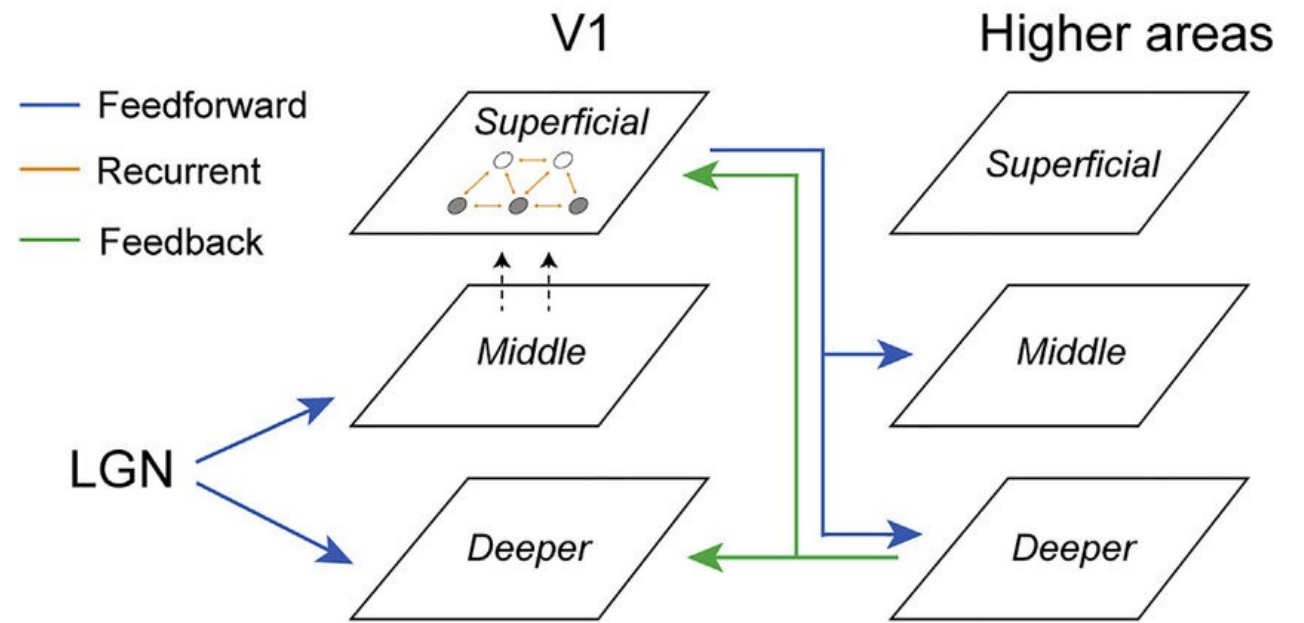
Internal models need:



Primate Brains

White Matter and Recurrent Connectivity Expanded

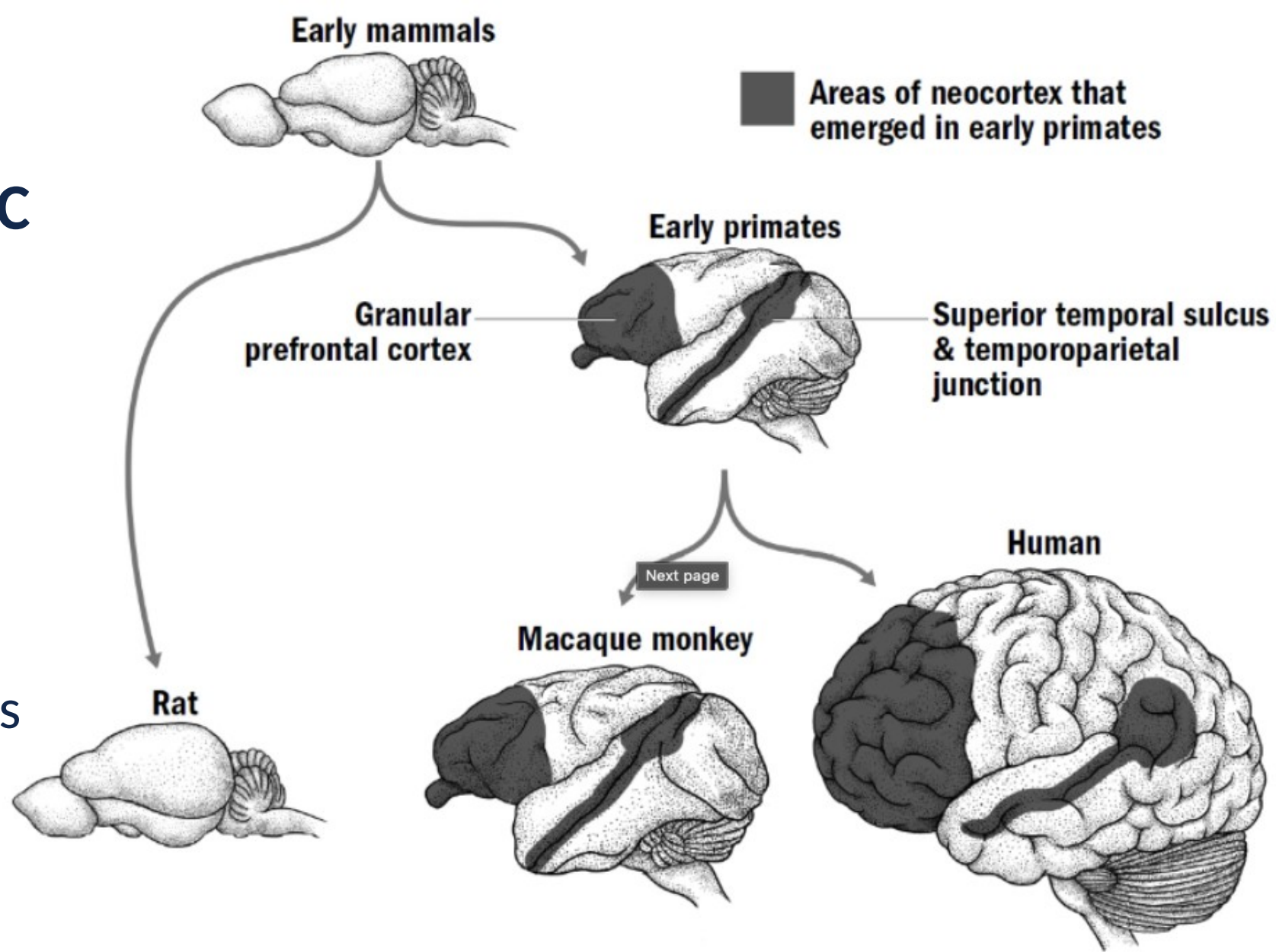
- Long-range pathways expanded
- Recurrent loops strengthened
- Persistent network dynamics



Primate Brains

Association Cortex + PFC Formed a Coordinated Workspace System

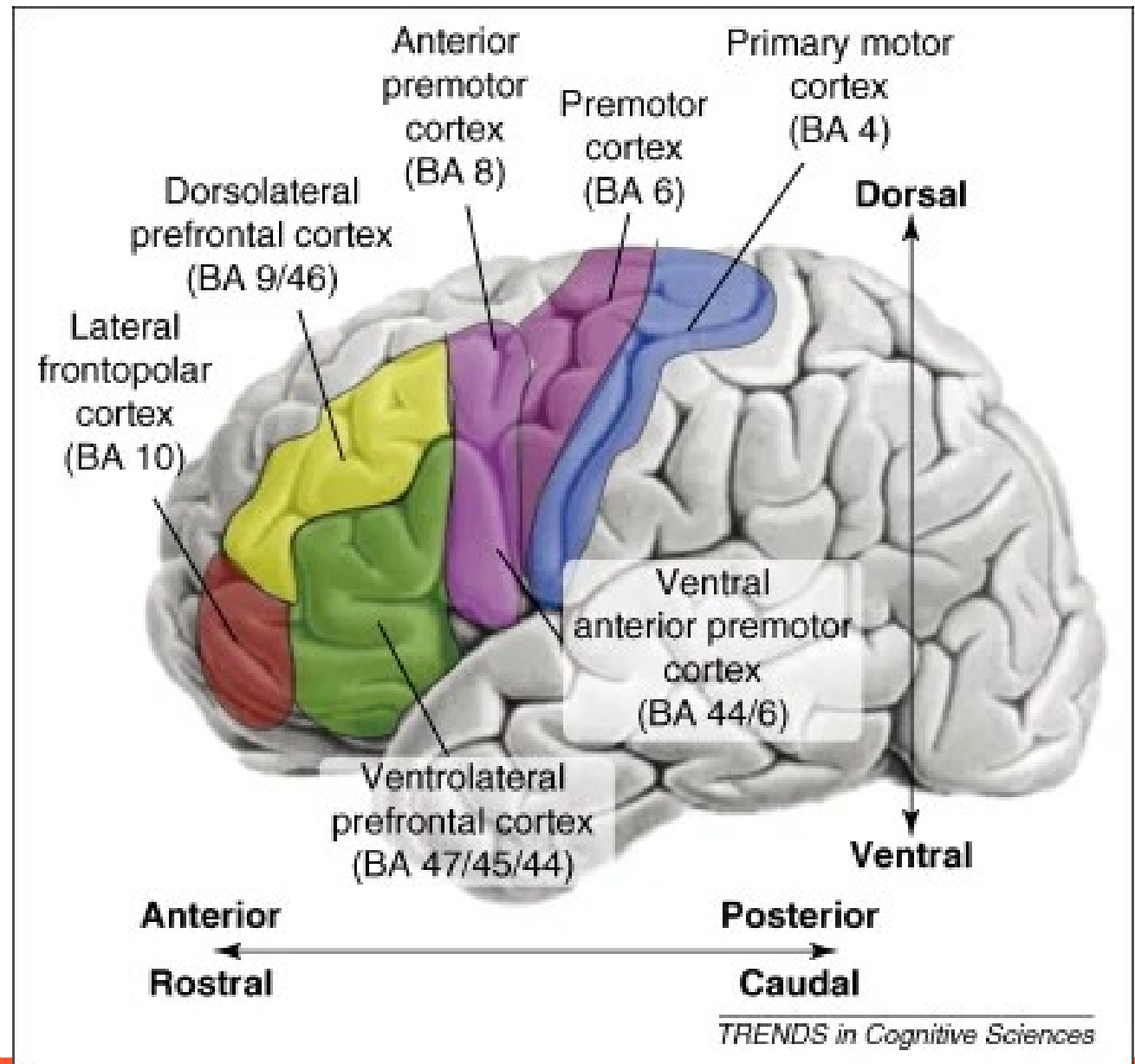
- Expanded association cortex
- PFC orchestrates use & routing
- Shared workspace across systems



Primate Brains

DorsoLateral Prefrontal Cortex (DLPFC): Task Sets & Counterfactual Rules

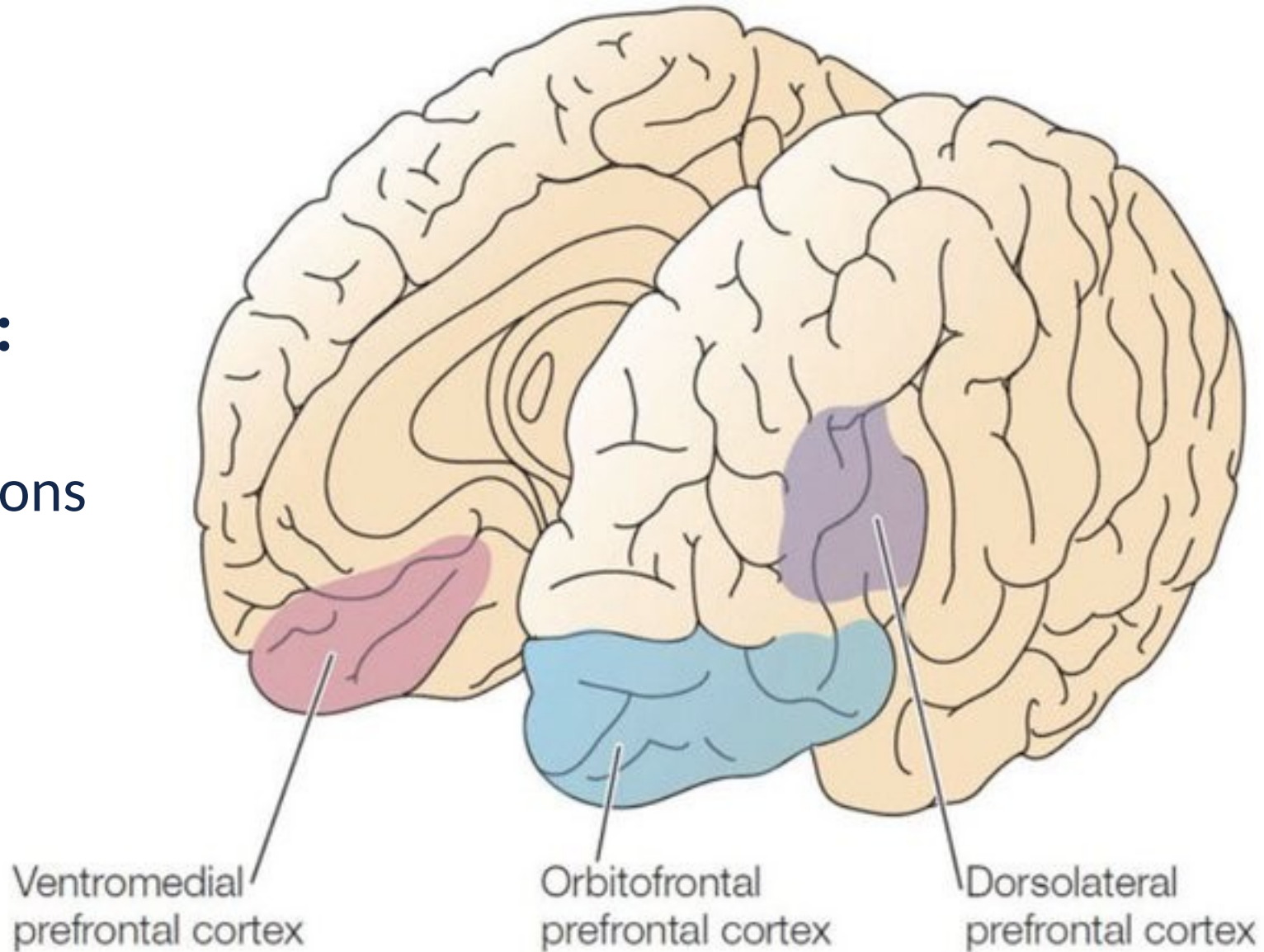
- Maintain task structure
- Evaluate “if-then” rules
- Support counterfactuals



Primate Brains

Ventromedial Prefrontal Cortex (VMPFC) / Orbitofrontal Cortex (OFC): Social Value & Norms

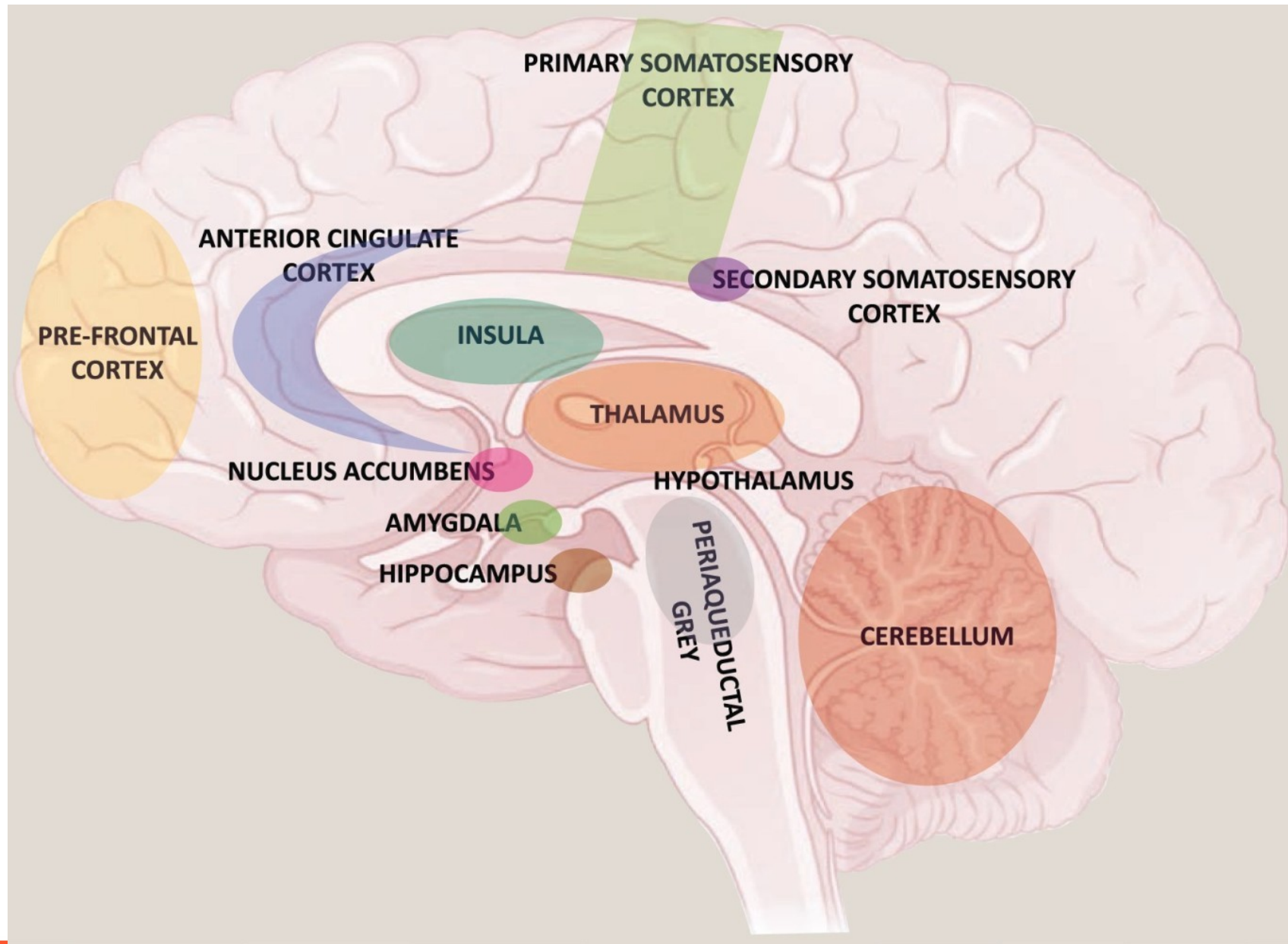
- Evaluate outcomes & intentions
- Track norms & social value
- Flexible, context-dependent valuation



Primate Brains

Anterior Cingulate Cortex (ACC): Social Error & Conflict Monitoring

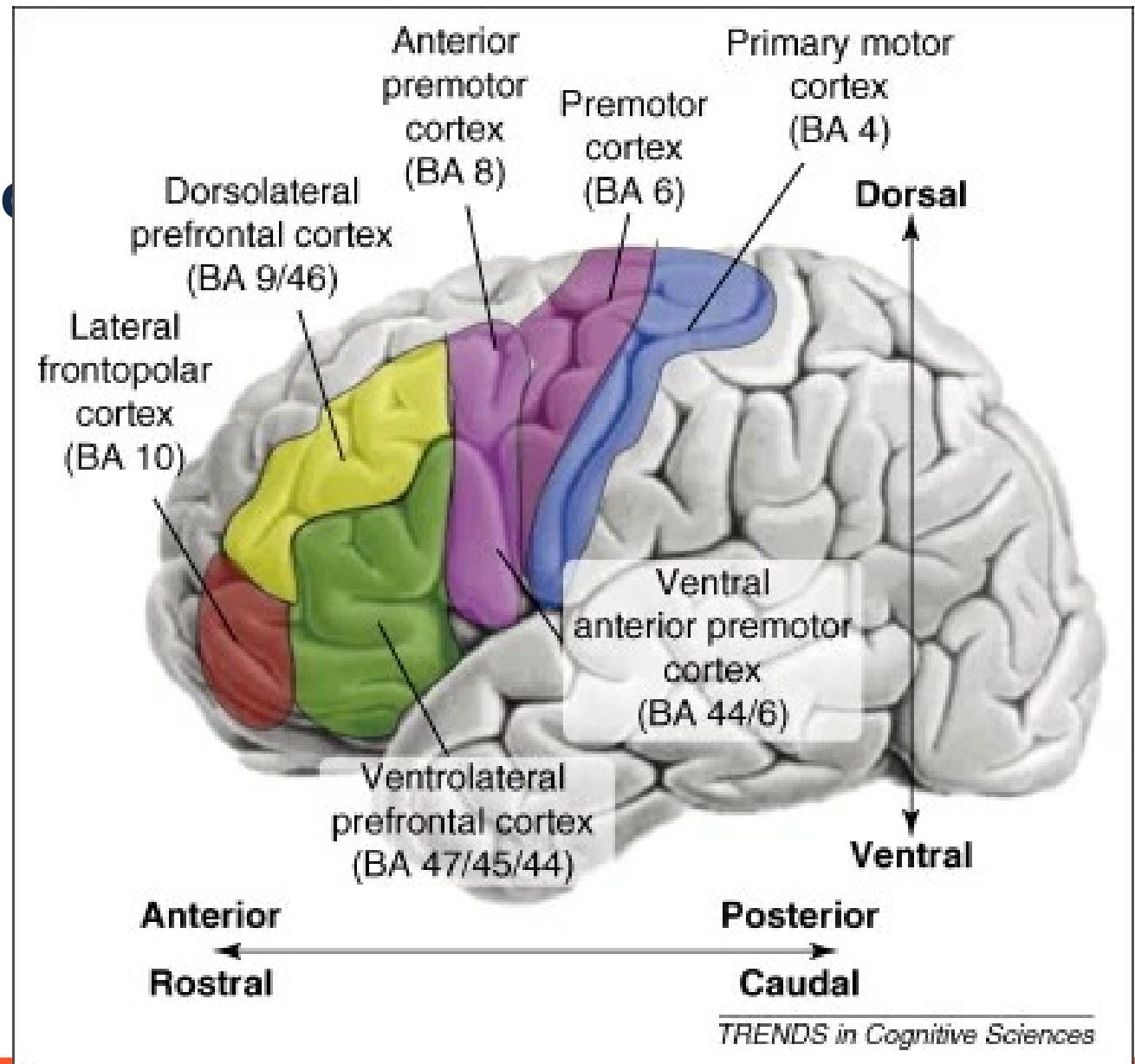
- Detect conflict & social errors
- Monitor outcomes vs expectations
- Adjust behavior when stakes change



Primate Brains

Frontopolar / BA10: Nested Goals & Strategy

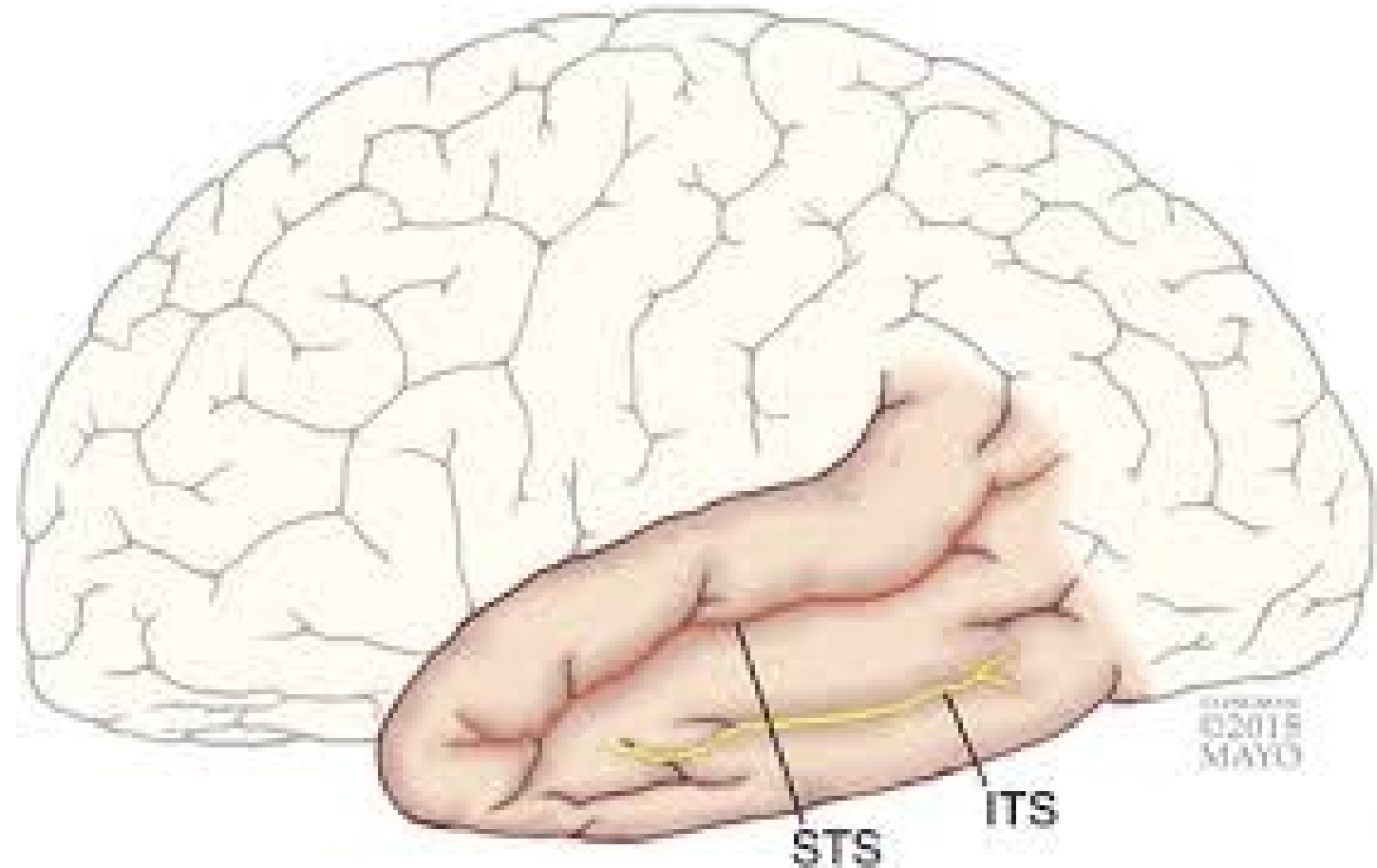
- Track multiple future goals
- Switch between plans
- Strategic depth & abstraction



Primate Brains

Temporal Cortex / STS: Social Perception Hub

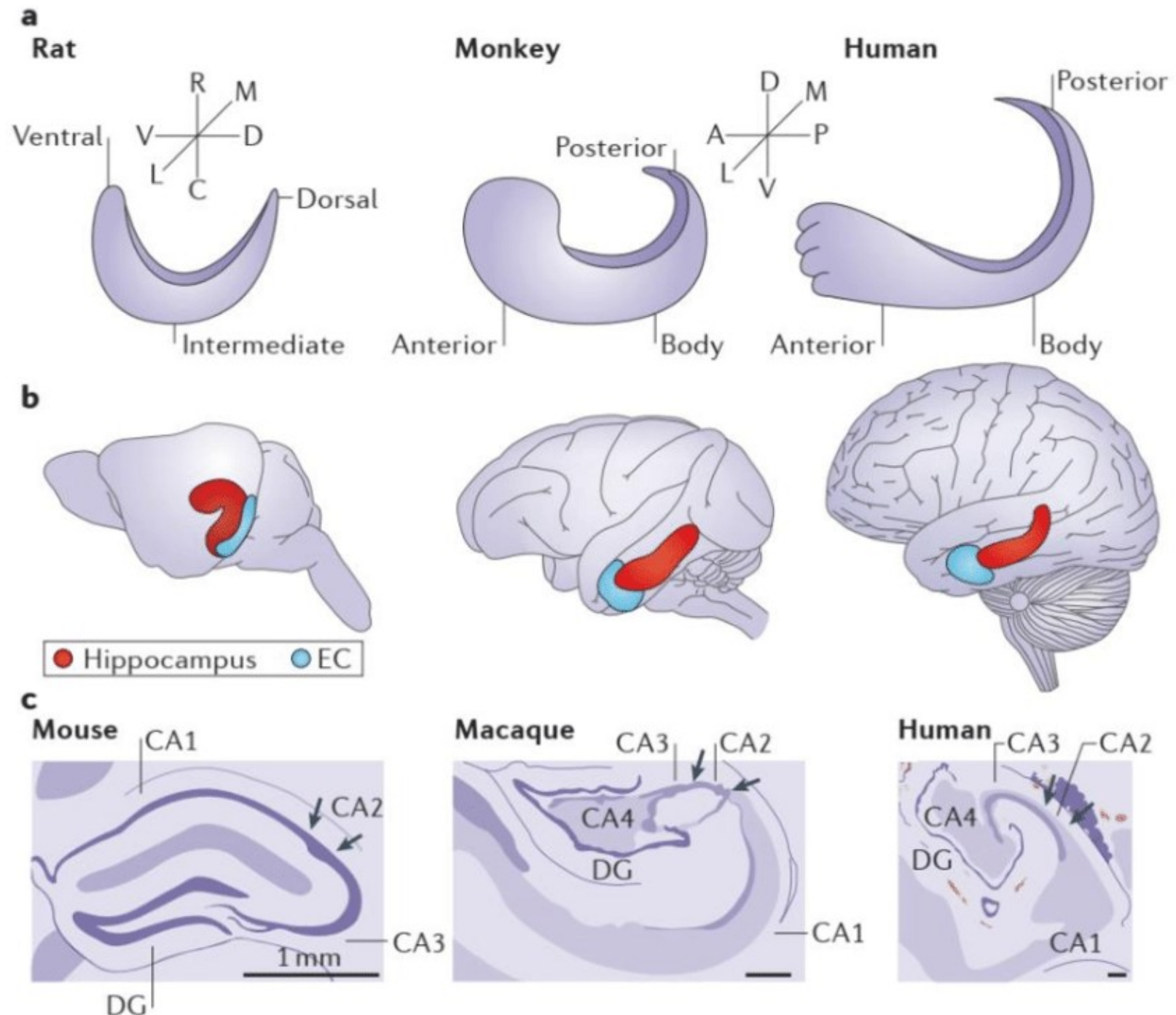
- Recognize individuals
- Read actions & expressions
- Social meaning from perception



Primate Brains

Hippocampus & Social Memory

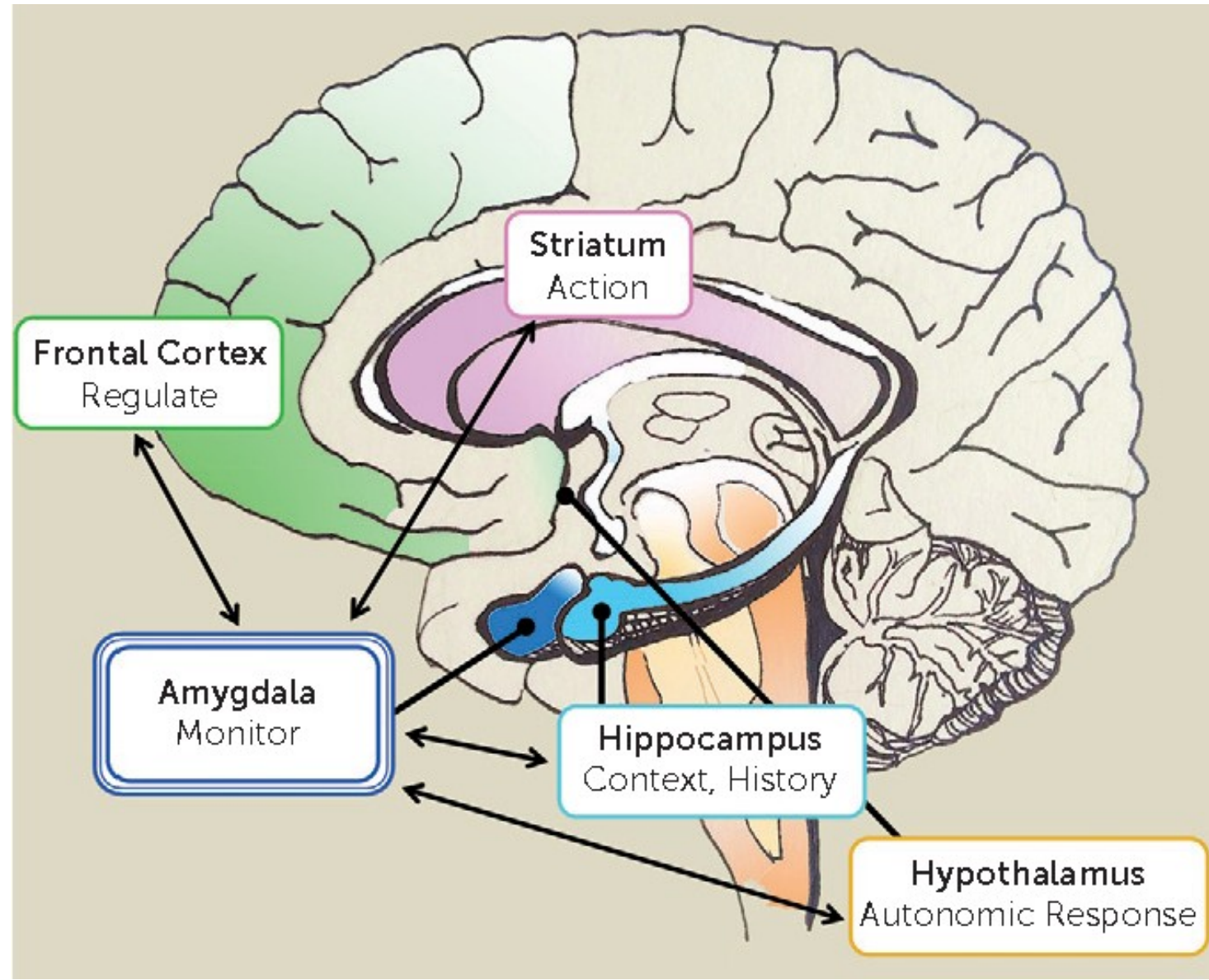
- Previously: episodic memory & spatial maps
- In primates: memory for **relationships, alliances, betrayals**
- Integrates social history into future strategy



Primate Brains

Basal Ganglia & Amygdala: Motivation, Status, Emotion

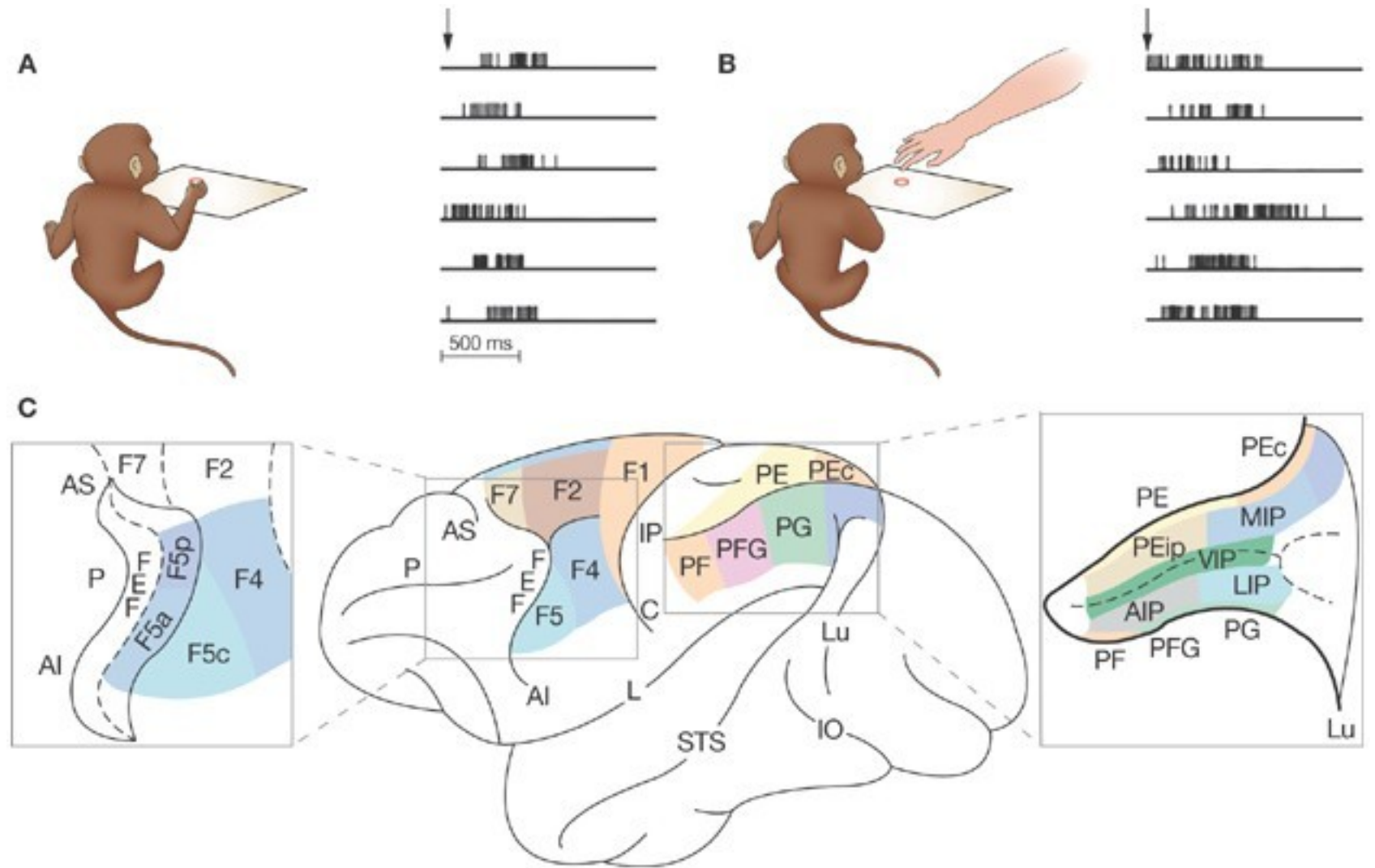
- Value learning for social life
- Status tracking, threat detection, bonding
- Primates: sensitivity to rank, reciprocity, reputation



Primate Brains

Mirror-Adjacent Action Networks: Imitation Substrate

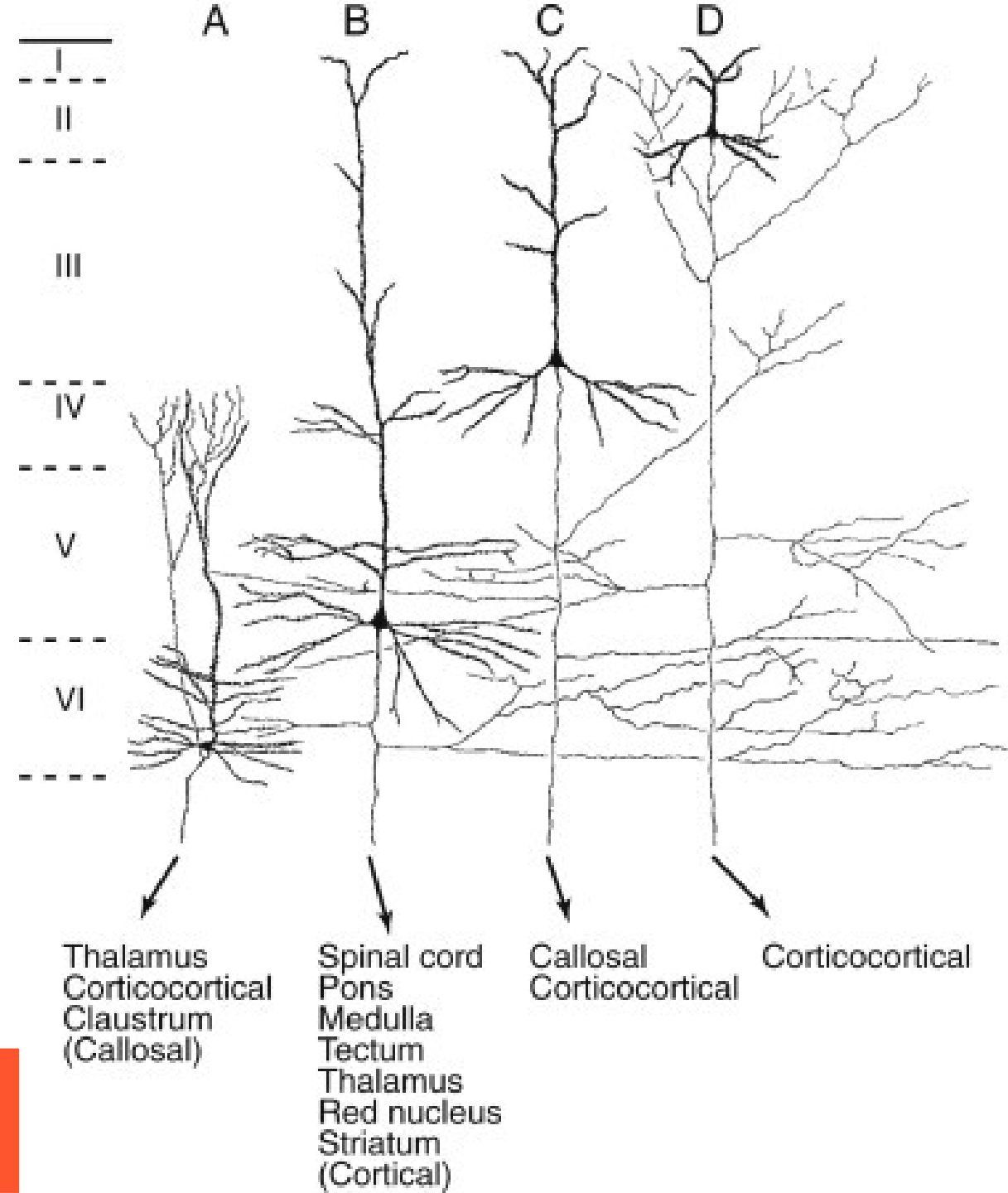
- Value learning for social life
- Status tracking, threat detection, bonding
- Primates: sensitivity to rank, reciprocity, reputation



Primate Brains

Expanded Layer III & V Pyramidal Neurons

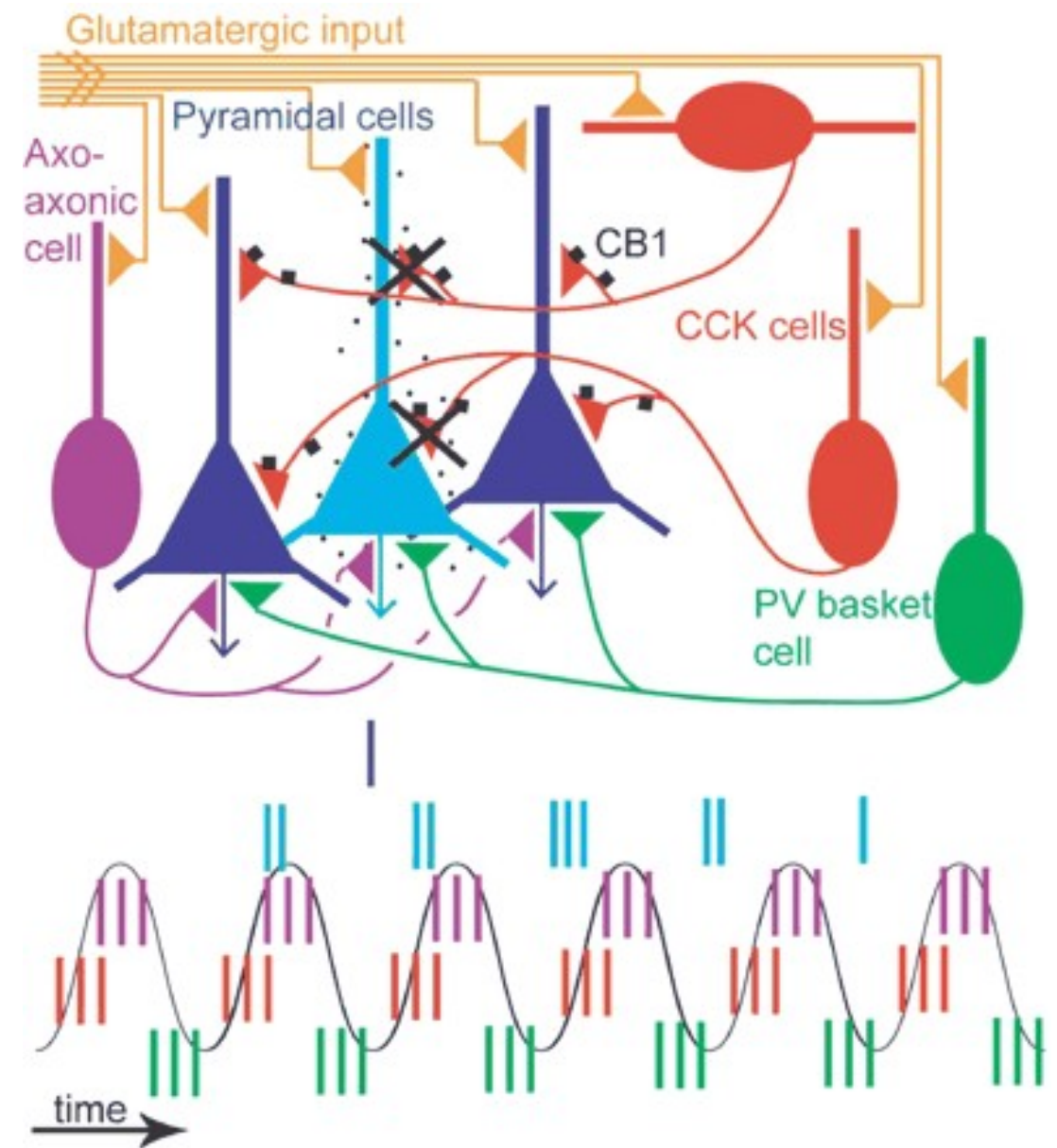
- Value learning for social life
- Status tracking, threat detection, bonding
- Primates: sensitivity to rank, reciprocity, reputation



Primate Brains

Inhibitory Interneurons & Precision Control

- Parvalbumin (PV) basket cells
- Chandelier cells
- Gating sequences

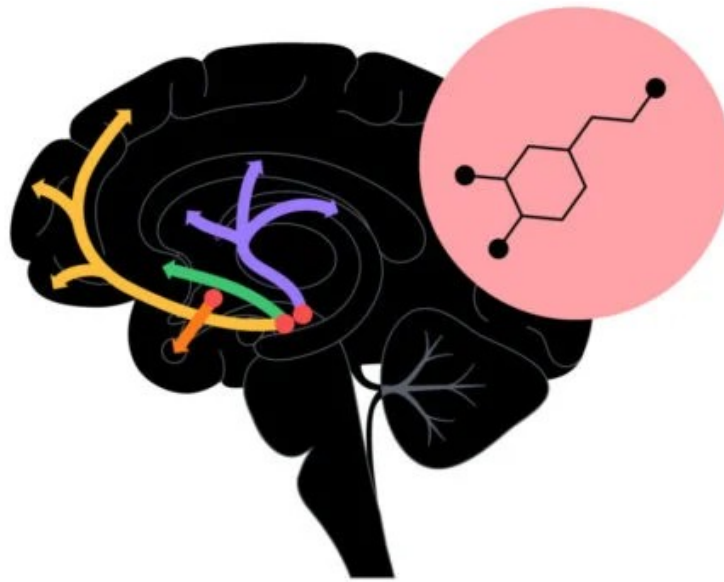


Primate Brains

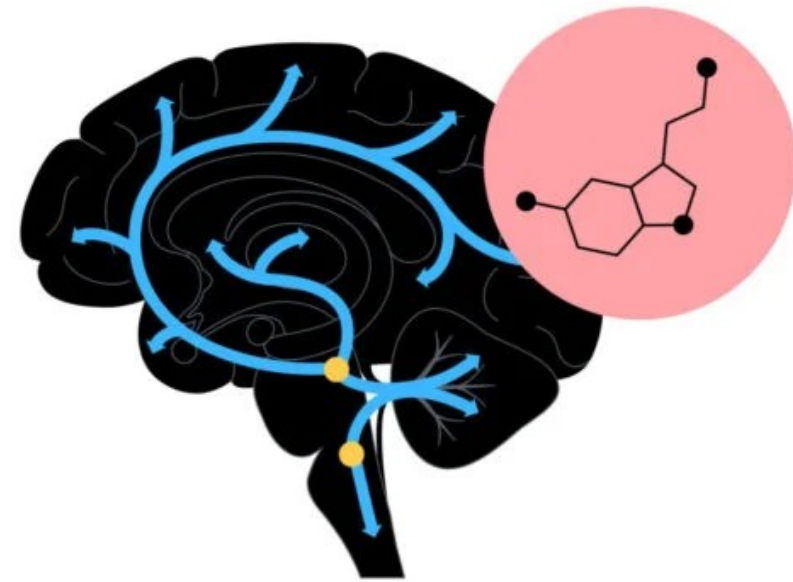
Neuromodulators Shaped Social Cognition

- **Dopamine:** value & future reward
- **Serotonin:** patience & impulse control
- **Oxytocin:** bonding & affiliation

DOPAMINE



SEROTONIN



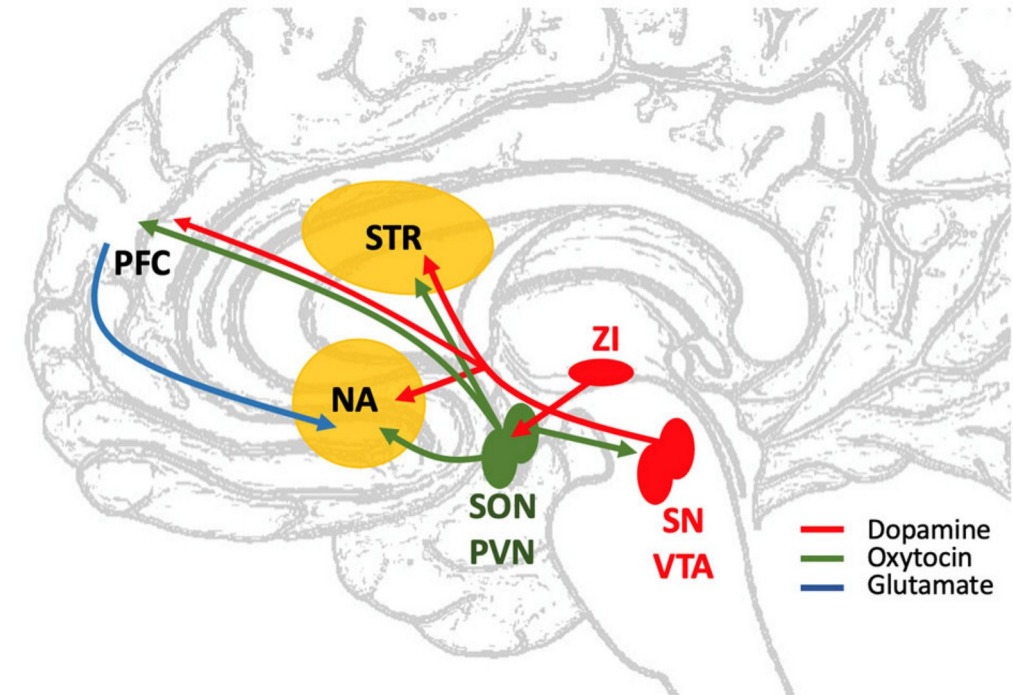
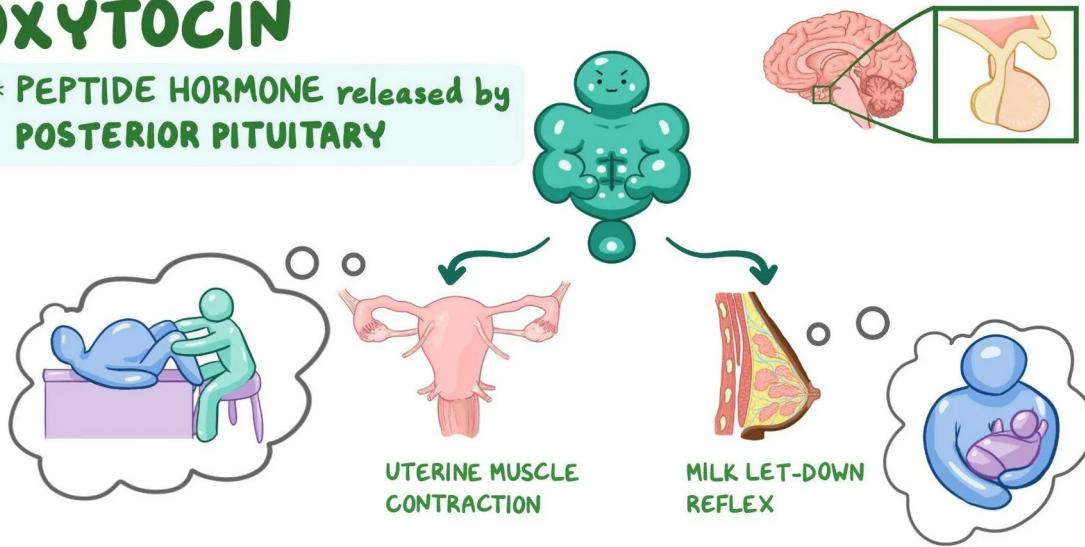
Primate Brains

Neuromodulators Shaped Social Cognition

- Dopamine: value & future reward
- Serotonin: patience & impulse control
- **Oxytocin**: bonding & affiliation

OXYTOCIN

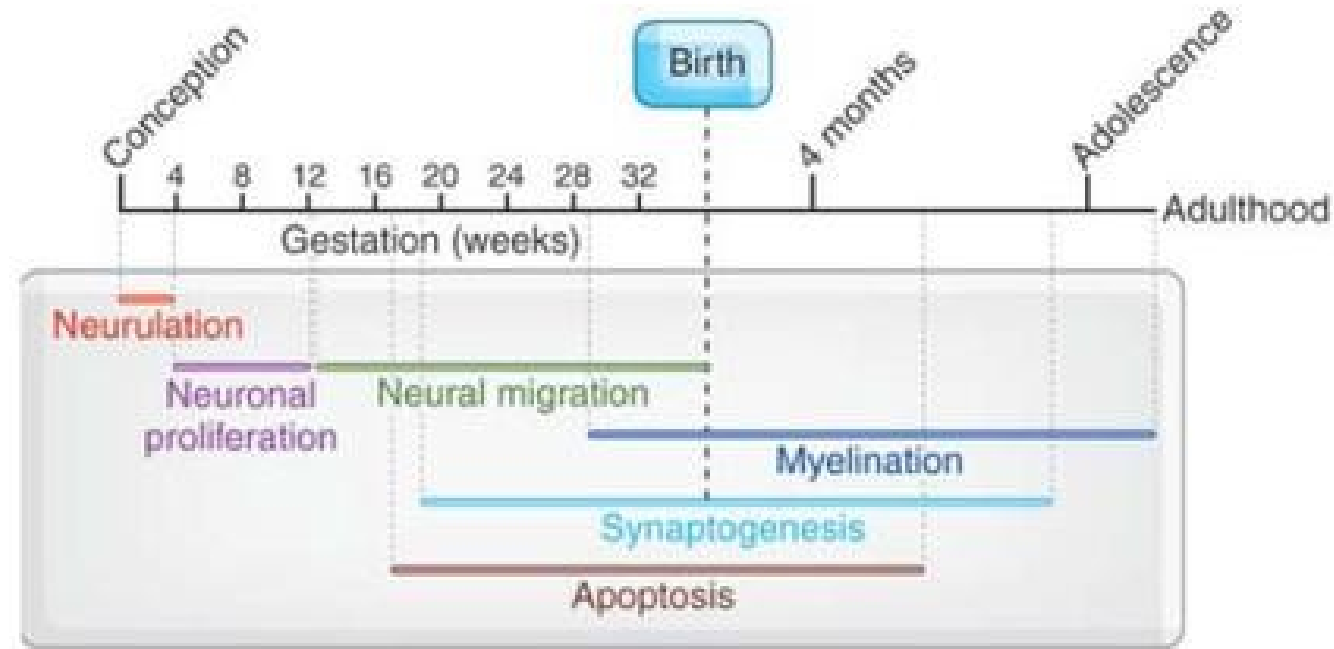
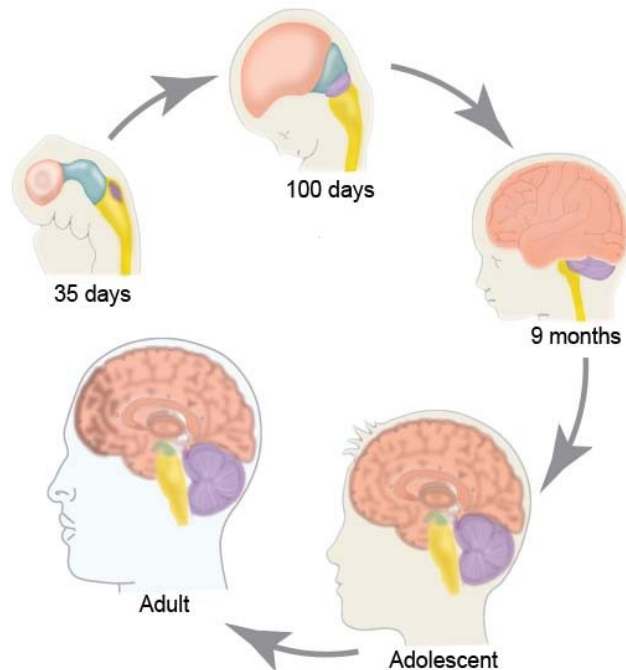
* PEPTIDE HORMONE released by
POSTERIOR PITUITARY



Primate Brains

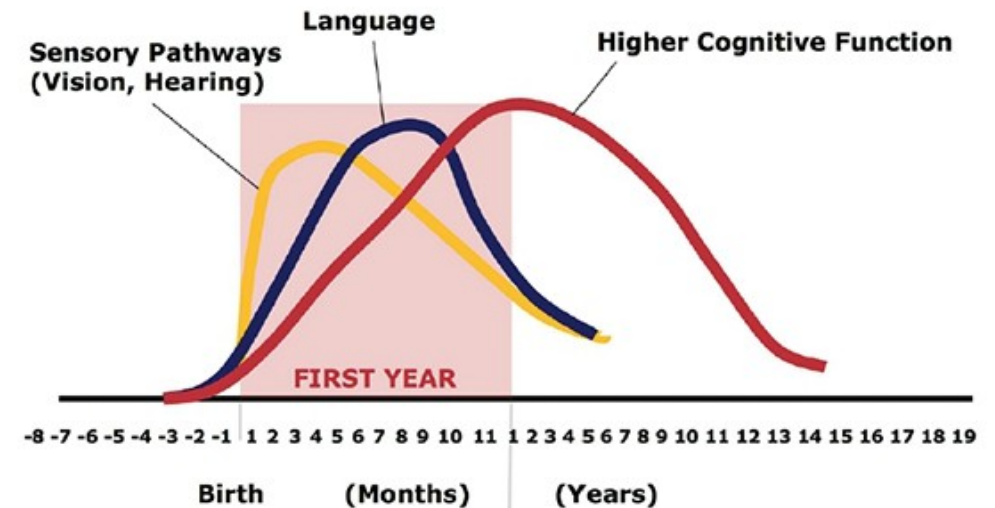
Development Built the Social Brain

- Long childhood
- Extended plasticity
- Experience-dependent tuning

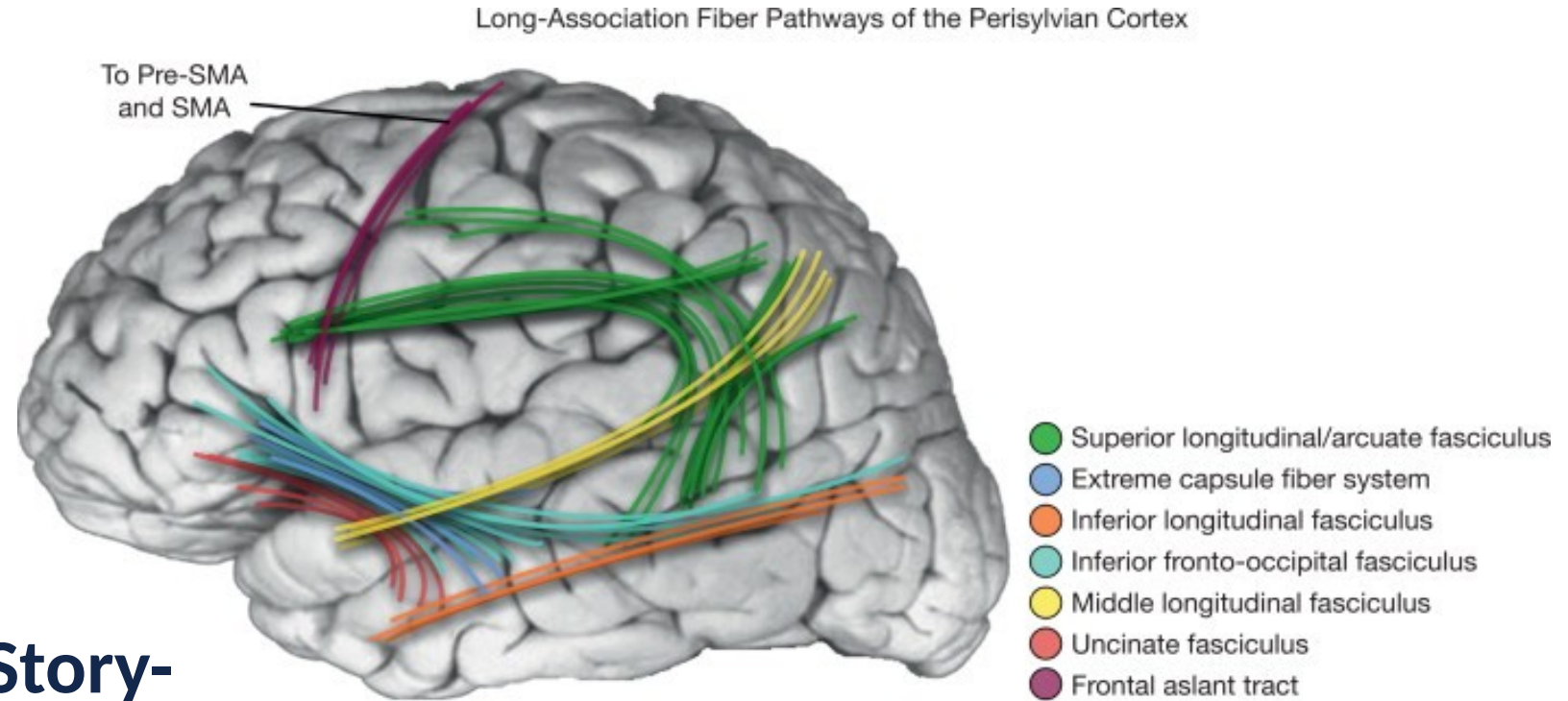


Human Brain Development

Neural Connections for Different Functions Develop Sequentially



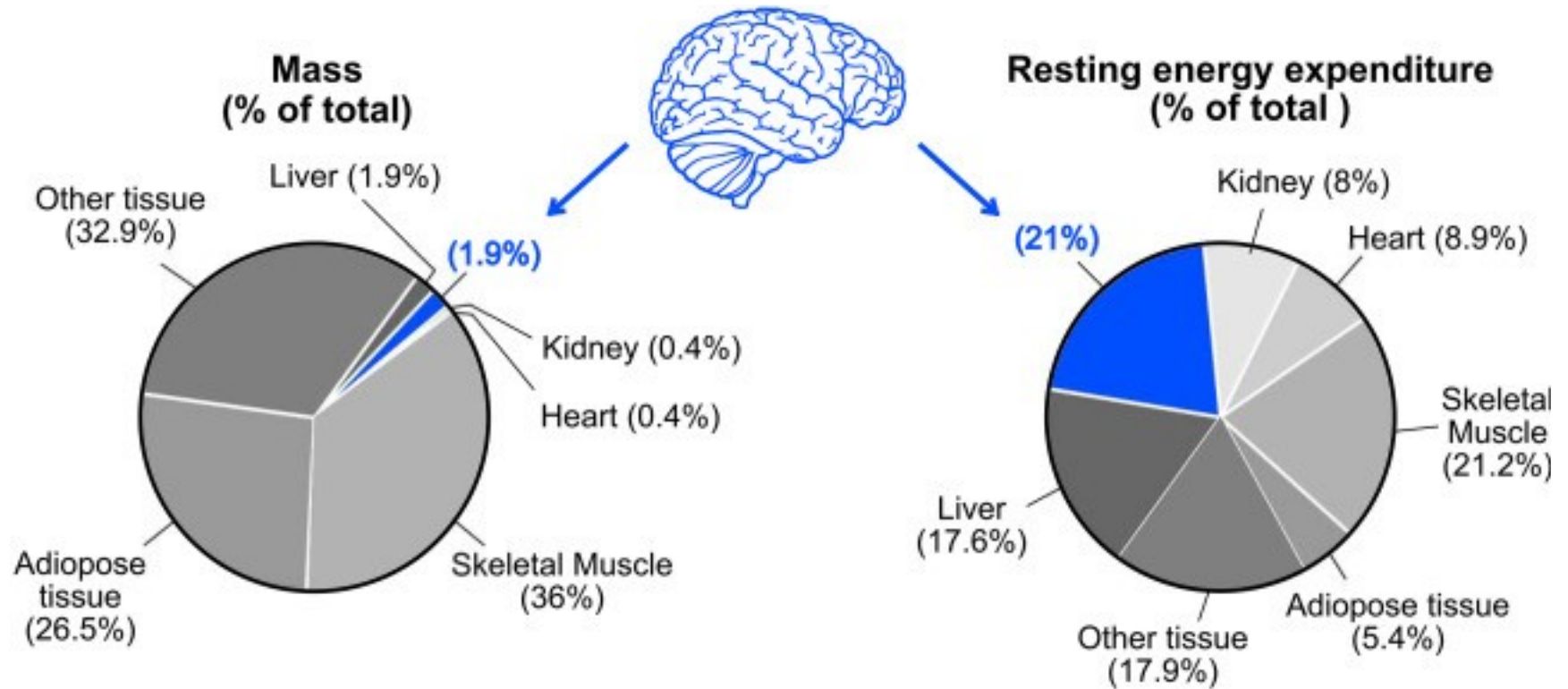
Primate Brains



White Matter Tracts for Story-Holding Minds

- Front-temporal highways
- Integrating memory + value + plans
- Keeping others “in mind”

Primate Brains



Energy Budget & Evolutionary Trade-offs

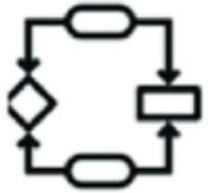
- Big brains = big costs
- Slow life histories
- Social strategy paid for it

Primate Brains



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Summary

- Social cognition across Marr's levels
- Today: neural architecture for social minds

Next Time

- Human brain evolution — language → symbolic thought